Muhammad Rommel Abbas

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Seeking Summer 2027 Internships: ML / Computer Vision / Software Development / Backend Engineering

EDUCATION

Rochester Institute of Technology — Expected May 2030
B.S. in Computer Science — Rochester, NY

TECHNICAL SKILLS

Languages: Python, Java, SQL, Bash
ML / Vision: PyTorch, scikit-learn, XGBoost, pandas, Ultralytics YOLO, OpenCV, VLMs, face recognition
LLMs / Speech: RAG, GraphRAG, LangChain, FAISS, Ollama, llama.cpp, CUDA, faster-whisper, Coqui XTTS, speaker ID
Backend / Systems: Django, FastAPI, PostgreSQL, SQLite, Neo4j, Docker, Linux, Nginx, Git, GitHub Actions, pytest
Robotics / Embedded: ROS 2, Nav2, Raspberry Pi, ESP32, Intel RealSense, motor control

RESEARCH

Undergraduate Researcher | Prof. John Thomas's Group, RIT — Oct 2026 – Present

- Sole member on Track 1 (EEG-to-Image) of the **NeurIPS 2026 EEG/EMG Foundation Challenge**; decoding images from EEG.
- Training on a local RTX 5080 with the lab's shared EEG preprocessing pipeline and 96 GB GPU server for larger runs.

EXPERIENCE

Machine Learning / Computer Vision Intern — Jun–Aug 2025
Punjab Safe Cities Authority | Government of Punjab — Lahore, Pakistan

- Trained YOLO models to detect urban safety hazards, including open manholes; annotated and preprocessed image/video datasets for a government authority operating **12,000+ cameras**.
- Developed an event-triggered encoder-to-VLM pipeline for camera analysis, invoking the vision-language model only on detected events to limit redundant inference.
- Implemented a multi-stage face-recognition pipeline for identity matching in camera imagery.

Cybersecurity Intern — Aug–Sep 2025
Counter Terrorism Department (CTD) | Government of KP — Peshawar, Pakistan

- Conducted open-source intelligence (OSINT) analysis of extremist activity to support threat assessments for public gatherings; presented findings to senior analysts.
- Awarded a **Class II Certificate of Commendation** by the Additional Inspector General of Police, KP.

PROJECTS

Regional Car Price Prediction — GitHub

- Built a used-car valuation application; developed a resumable scraper for **59,668 PakWheels listings** and curated a deduplicated 48K-row dataset.
- Trained XGBoost with a leakage-aware grouped split, achieving **0.835 R^2 and 11.5% MAPE** on a 9,674-row holdout set.
- Served predictions through Django; implemented 20 tests with 81% coverage and GitHub Actions CI across Python 3.11–3.13.

ALIRA – Local Voice AI Assistant & Robot — Dec 2024 – Present

- Developed a PyTorch wake-word detector, iterating from an LSTM to a 3-layer CNN on 80-bin Mel spectrograms; packaged as a Python library.
- Built a speech-processing pipeline with Silero VAD and DeepFilterNet3; benchmarked emotion2vec inference at **0.05 real-time factor** on CPU.
- Developing person-specific Neo4j GraphRAG with graph traversal and vector retrieval; integrating face recognition and speaker identification to link users to conversation history.
- Developing a ROS 2 mobile robot with Raspberry Pi audio capture, GPU-based inference, and encoder odometry; planning Nav2 navigation with RealSense depth sensing.

SkillTree – Tech Job Market Intelligence — skilltree.handoverit.com

- Built a Django platform aggregating **69,491 job postings from 44+ sources** to analyze technical skill demand.
- Extracted and normalized skills using Llama models via Groq; stored results in SQLite for trend queries and self-hosted on Linux with Gunicorn, Nginx, and Cloudflare Tunnel.

Additional projects
Local LLM Inference: Tuned llama.cpp/CUDA for Qwen3-Coder 30B on an RTX 5080 with MoE expert offloading to CPU.
Hermes AI agent (GitHub): Self-hosted a Discord/Matrix agent on Proxmox with an ESP32 status display.
WhisperKey (GitHub): Developed open-source offline push-to-talk dictation for Linux using faster-whisper.

LEADERSHIP & INVOLVEMENT

Computer Science House (CSH), RIT — Aug 2026 – Present

- Selected as an off-floor member; built a read-only Slack bot that ingests channel history into a local retrieval agent.
- Invited to present an R&D seminar on local LLM inference and voice-assistant architecture.

RITSEC (RIT Cybersecurity Club) — Aug 2026 – Present

- Active member; attend weekly technical meetings on offensive security and CTF tooling.